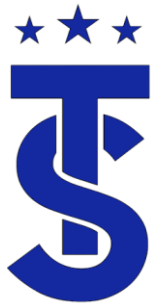




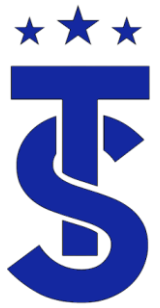
FW Webb Training Exercises

DVM S Refrigeration Circuits and Troubleshooting



Overview

- Use the operational data for each mode to troubleshoot each exercise.
- State your diagnosis and how you came to your decision.
- Explain how the system responds to the changes in the system.



Test Mode Operation

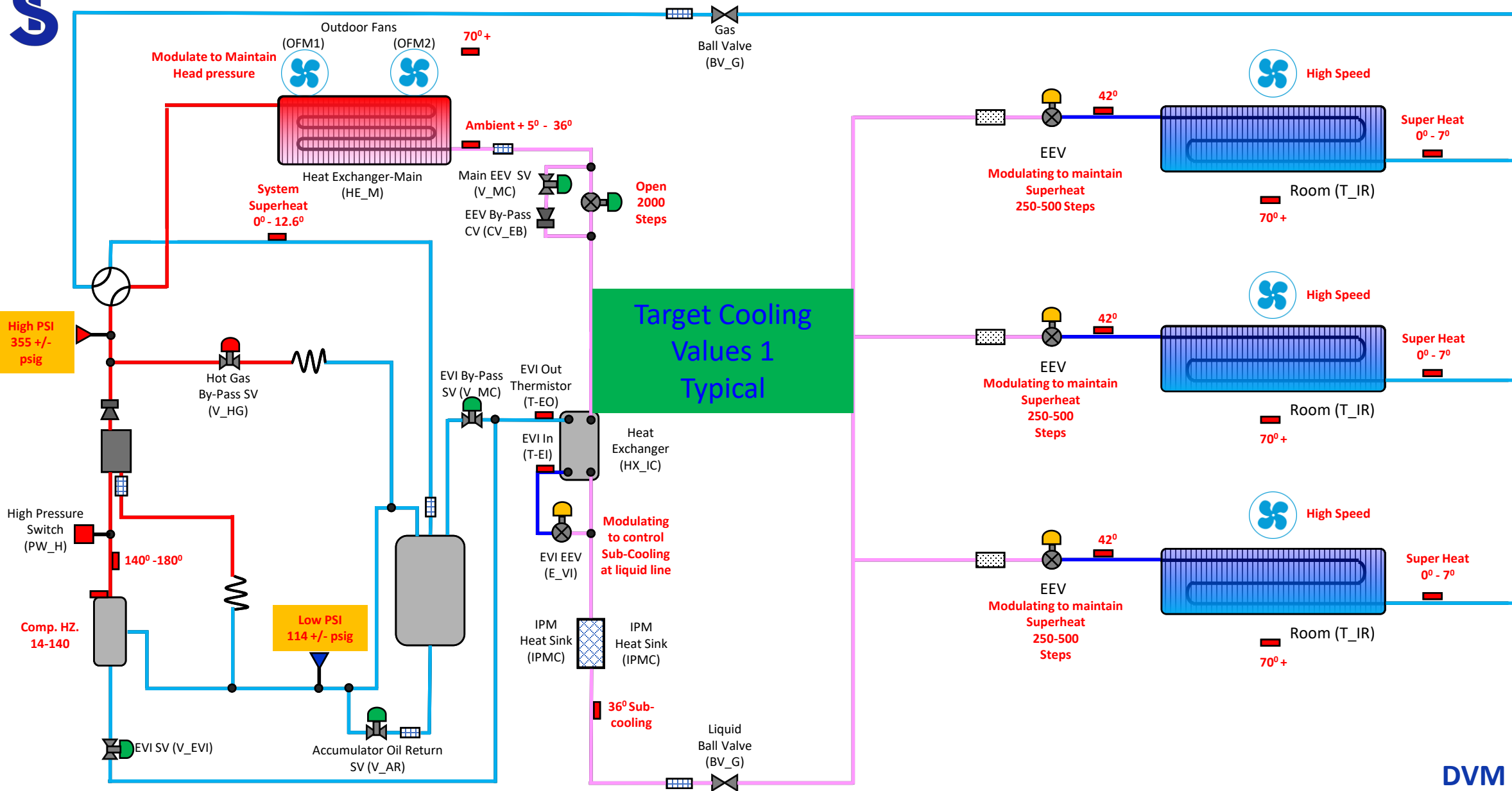
- Outdoor Unit Operation
 - Compressor hertz (These values can vary based on ODU)
 - Normal Cooling Max 140 hz
 - Test Mode Cooling operation 80 hz
 - Normal Heating Max 160 hz.
 - Test Mode Heating operation 90hz
 - In test mode the all-other controls operate as normal to maintain target operation.
- Indoor units
 - Fan motor: high speed
 - Set Point Temperature (100% demand)
 - 3 degree(Celsius) for colling 36⁰ F
 - 40 degree(Celsius) for heating 104⁰ F
 - EEV operates as normal to maintain target operation
- Protection logic
 - Follows normal operation



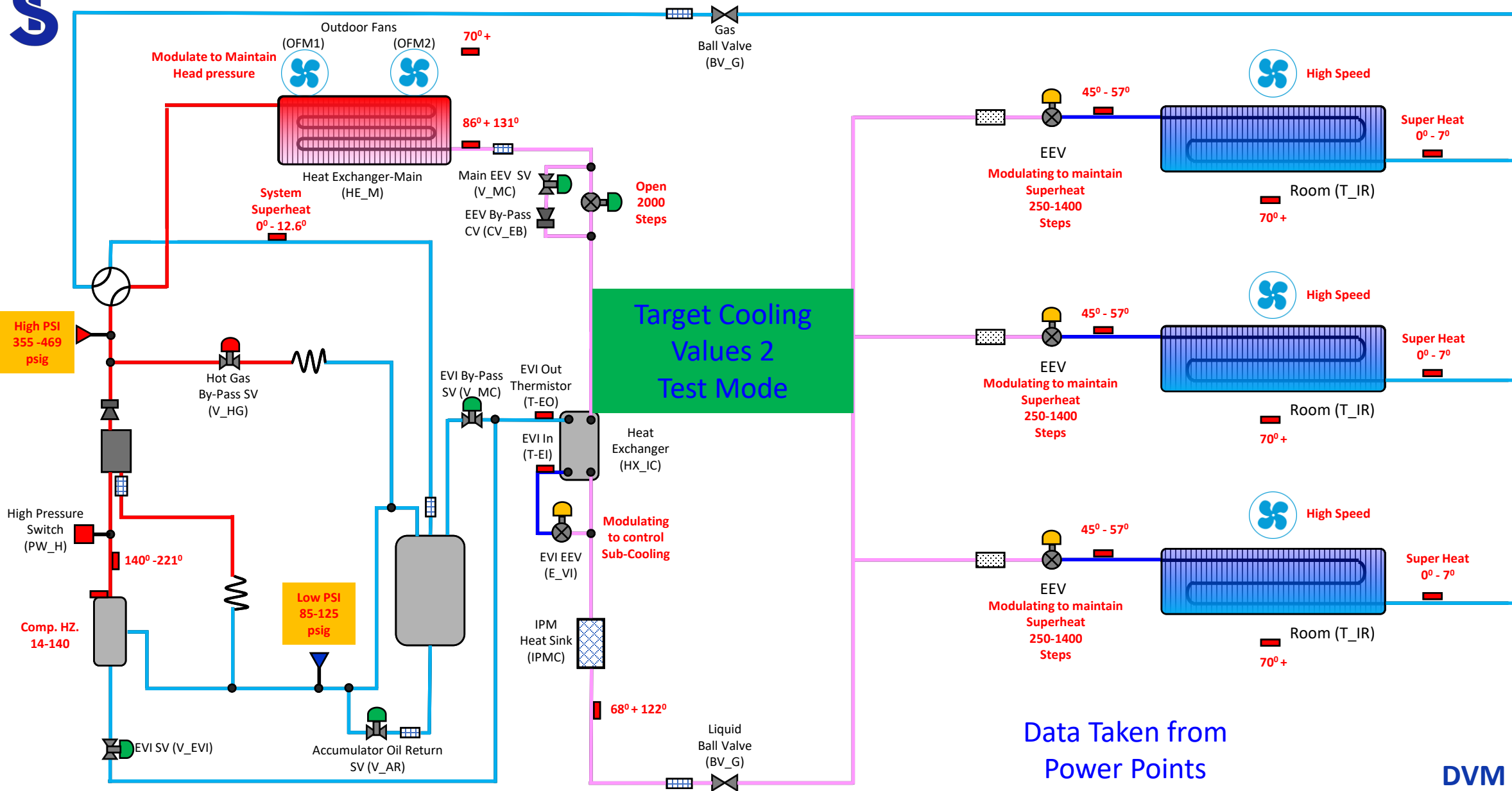
Check point of cooling trial operation

※ The data is not absolute one, only can be used to check the abnormal status

Item	Normal	Check point
EVA IN Temp.	44.6° - 57.2°	▷ Check target Eva in temperature of option setting(cooling capacity correction) The default : EVA IN 44.6°-57.2°, Super heat = EVA OUT - EVA IN = 1.8° – 7.2° (Normal) ▷ Refrigerant shortage check With all IDU operation, If more than 50% of the indoor unit's SH is over 10.6 & EEV step of those unit is over 900 step . (In case of EEV RAC : 1300 step) ▷ Refrigerant over charged check With all IDU operation, as the room temp. is over 77°, If the majority of IDU of SH is 0 -1.8° & these EEV step is less then 200step . (In case of EEV RAC : 300 step) ▷ If only few indoor unit's EEV step is the Max. step, check the pipe length from the first branch pipe to indoor unit (limited 45m) also ensure that if there is any kind of flow resistance
EVA OUT Temp.	44.6° – 57.2°	
EEV Step of Indoor Unit	250 ~ 900	
Low Pressure psig	85 - 135	▷ The low pressure varies from 85 – 135 according to the piping length & target low pressure. When the ambient temperature is high or the indoor load is big, the low pressure may be higher. ▷ High pressure will rise when the ambient temperature rises (Max. 512 psig) When the ambient temperature decreases, high pressure may be decreased as well
High Pressure psig	355 - 469	
Compressor Discharge Temp.	140° - 221°	▷ High pressure is over 355 psig but discharge temp. is less than 140° (DSH less than 36°) Please check the EVI EEV, PHE leakage, indoor EEV opening error, Indoor fan operating error.
Condensing Temp.	95° - 131°	▷ Condensing temperature will be changed according to the high pressure
Liquid Tube Temp.	77° - 122°	▷ Condensing temp. – liquid tube temp.= Sub-cooling (9° - 36°)



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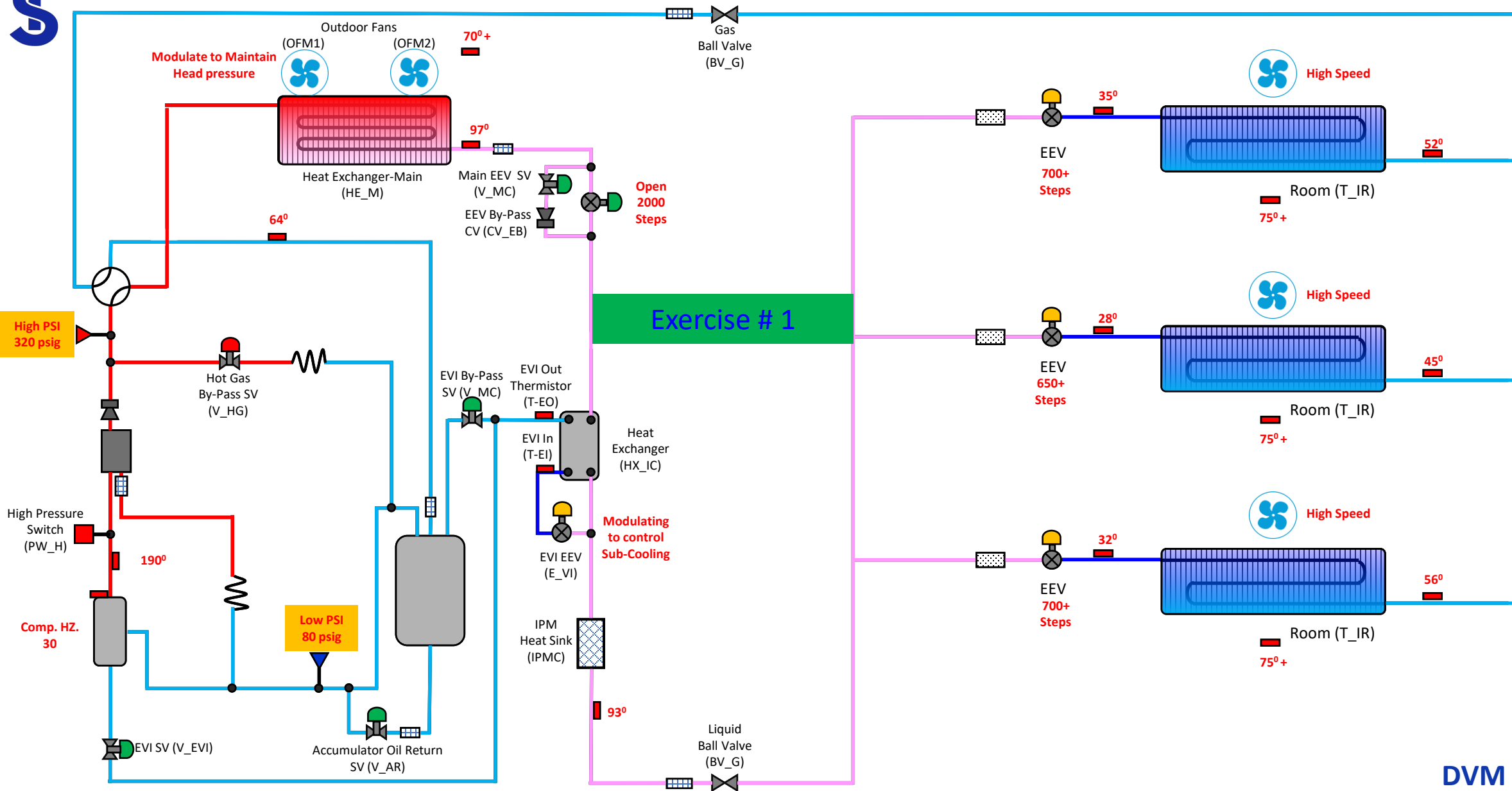
Exercise # 1

Operational Characteristics

- ☐ Inadequate cooling
- ☐ System Cools when the load is low but not when its high
- ☐ Compressor does not seem to run at a high enough speed for the load

We can use this section to put in operational characteristics of the system and any other data pertinent to the fault, like the compressor is in protection mode or system is throwing a fault. Thoughts





DVM S HP

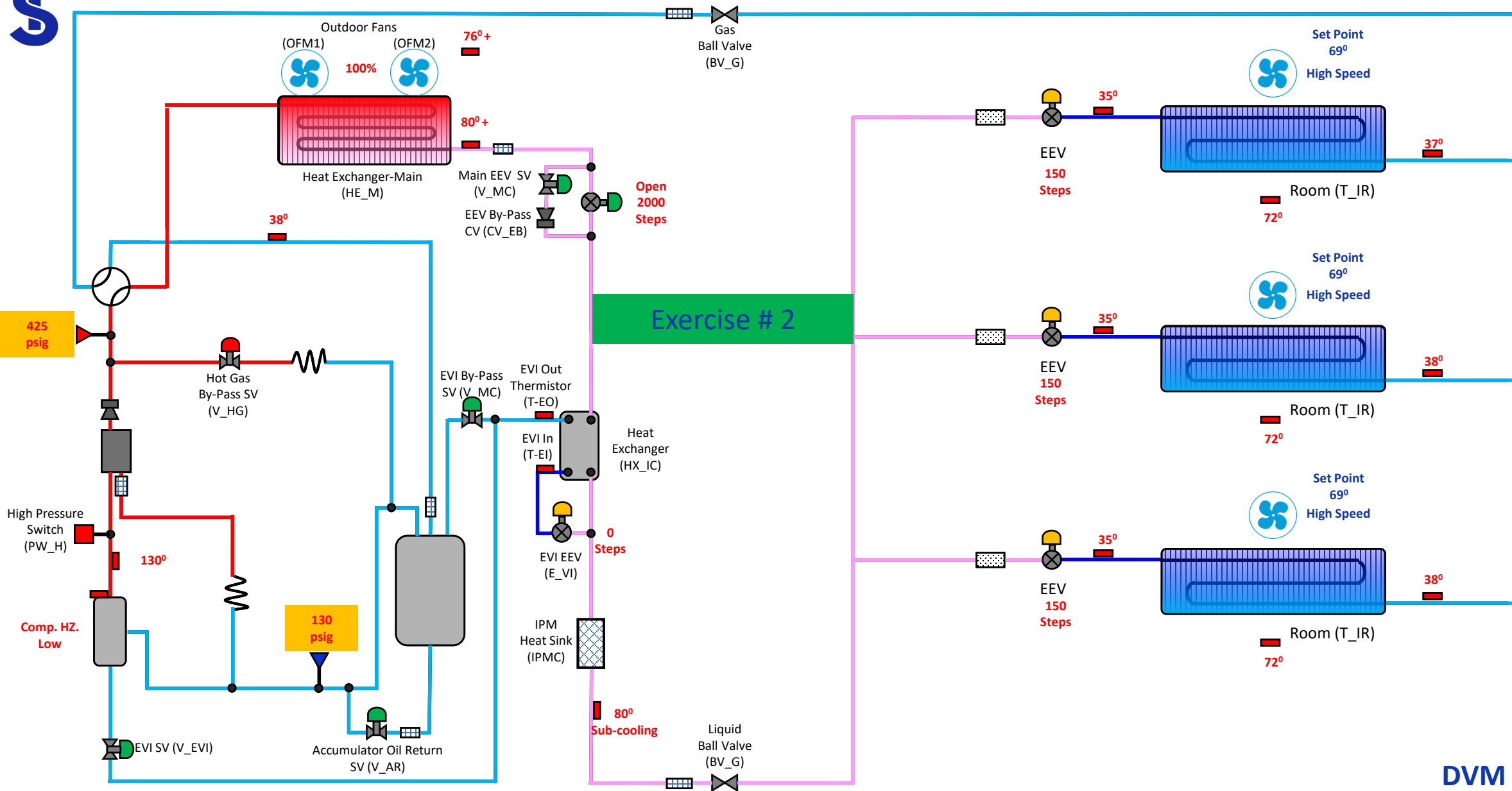
Exercise # 2

Operational Characteristics

- ☐ Inadequate cooling
- ☐ System Cools when the load is low but not when its high
- ☐ Compressor does not seem to run at a high enough speed for the load

We can use this section to put in operational characteristics of the system and any other data pertinent to the fault, like the compressor is in protection mode or system is throwing a fault. Thoughts





DVM S HP

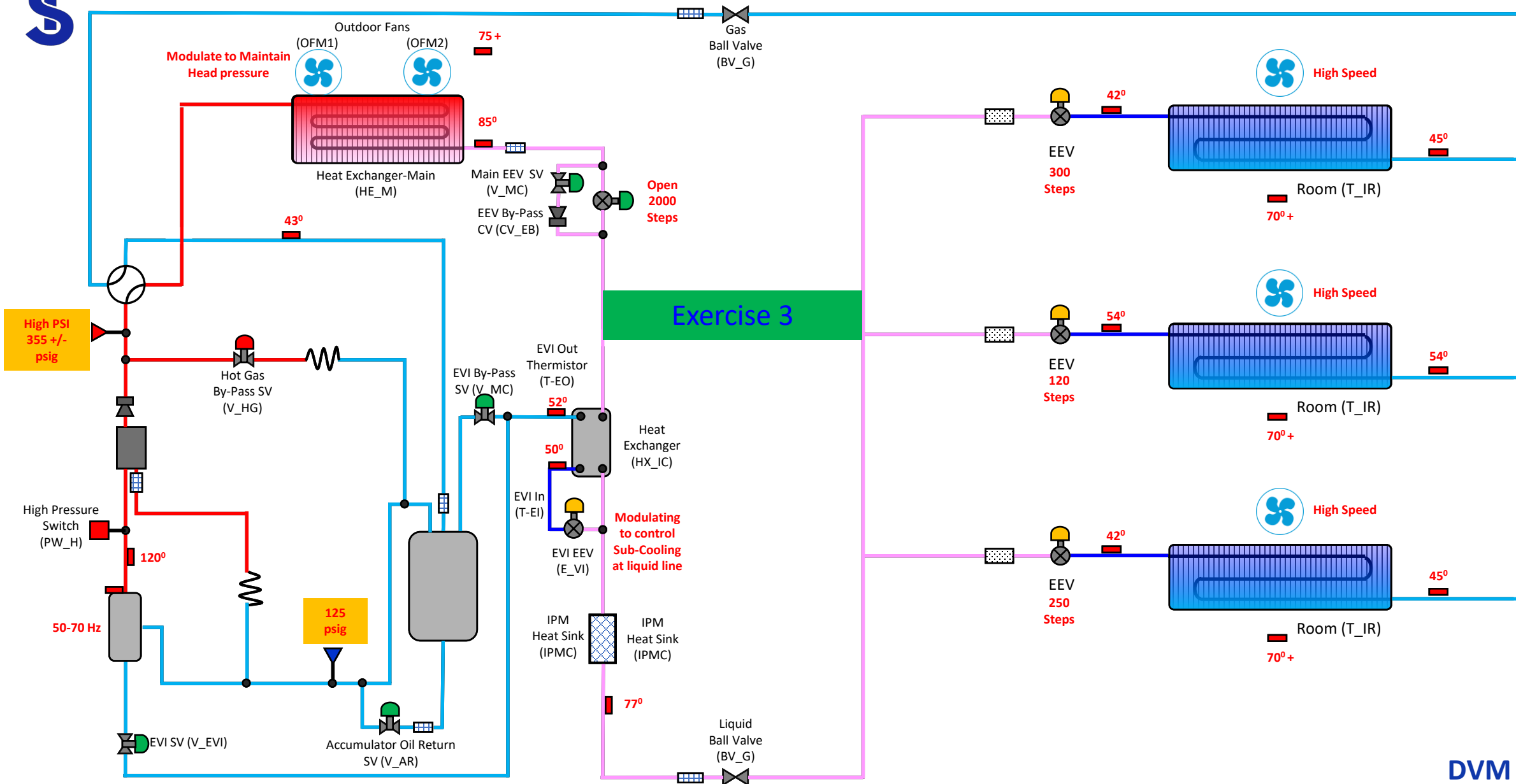
Exercise # 3

Operational Characteristics

- ☐ Zone 2 is not cooling
- ☐ Zones 1 & 3 have diminished cooling capacity

We can use this section to put in operational characteristics of the system and any other data pertinent to the fault, like the compressor is in protection mode or system is throwing a fault. Thoughts





DVM S HP

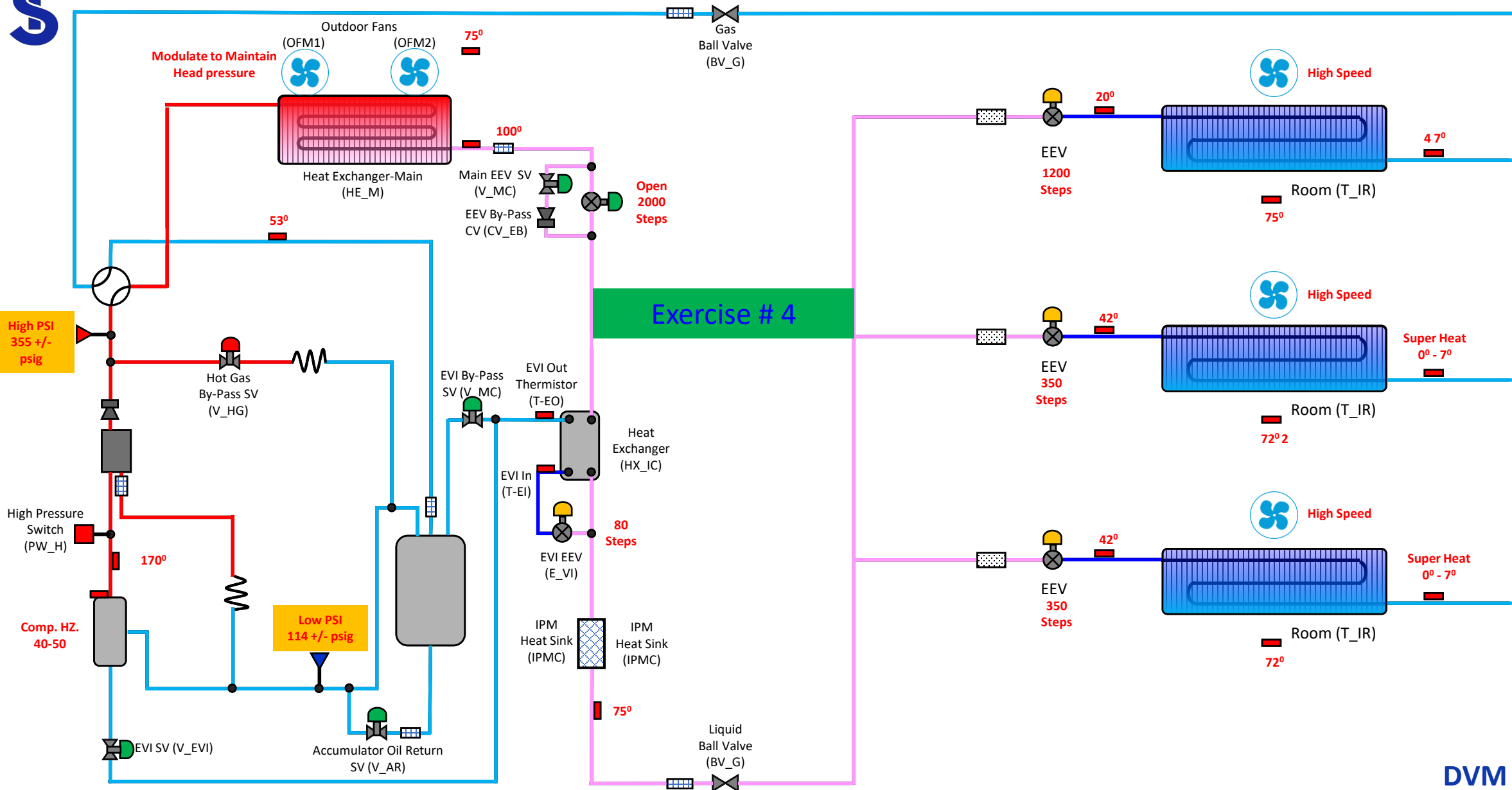
Exercise # 4

Operational Characteristics

- ☐ Zone 1 is not cooling
- ☐ Other zones are maintaining space temperature

We can use this section to put in operational characteristics of the system and any other data pertinent to the fault, like the compressor is in protection mode or system is throwing a fault. Thoughts





DVM S HP

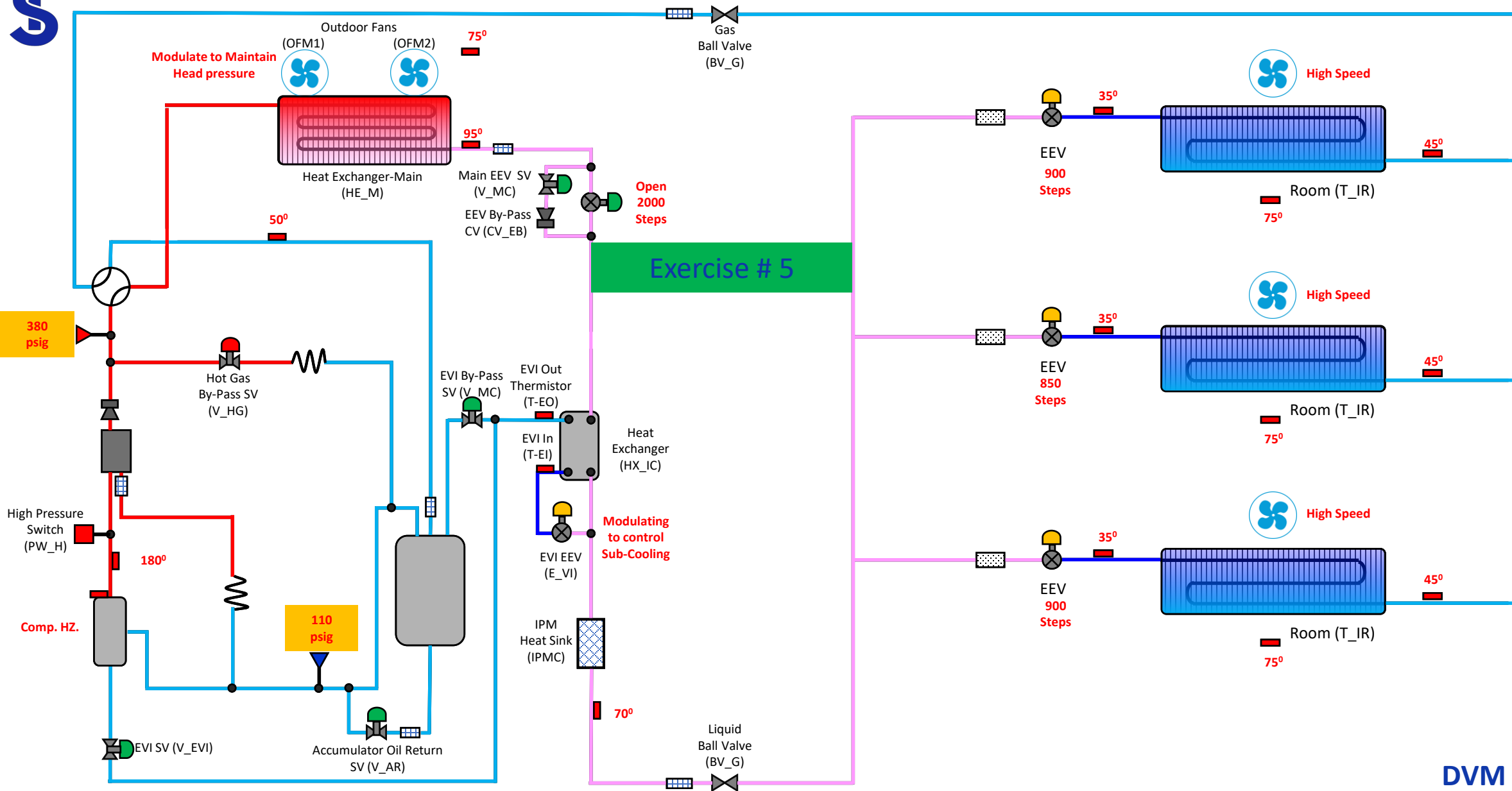
Exercise # 5

Operational Characteristics

- ☐ Inadequate cooling for all zones
- ☐ System Cools when the load is low but not when its high

We can use this section to put in operational characteristics of the system and any other data pertinent to the fault, like the compressor is in protection mode or system is throwing a fault. Thoughts





DVM S HP

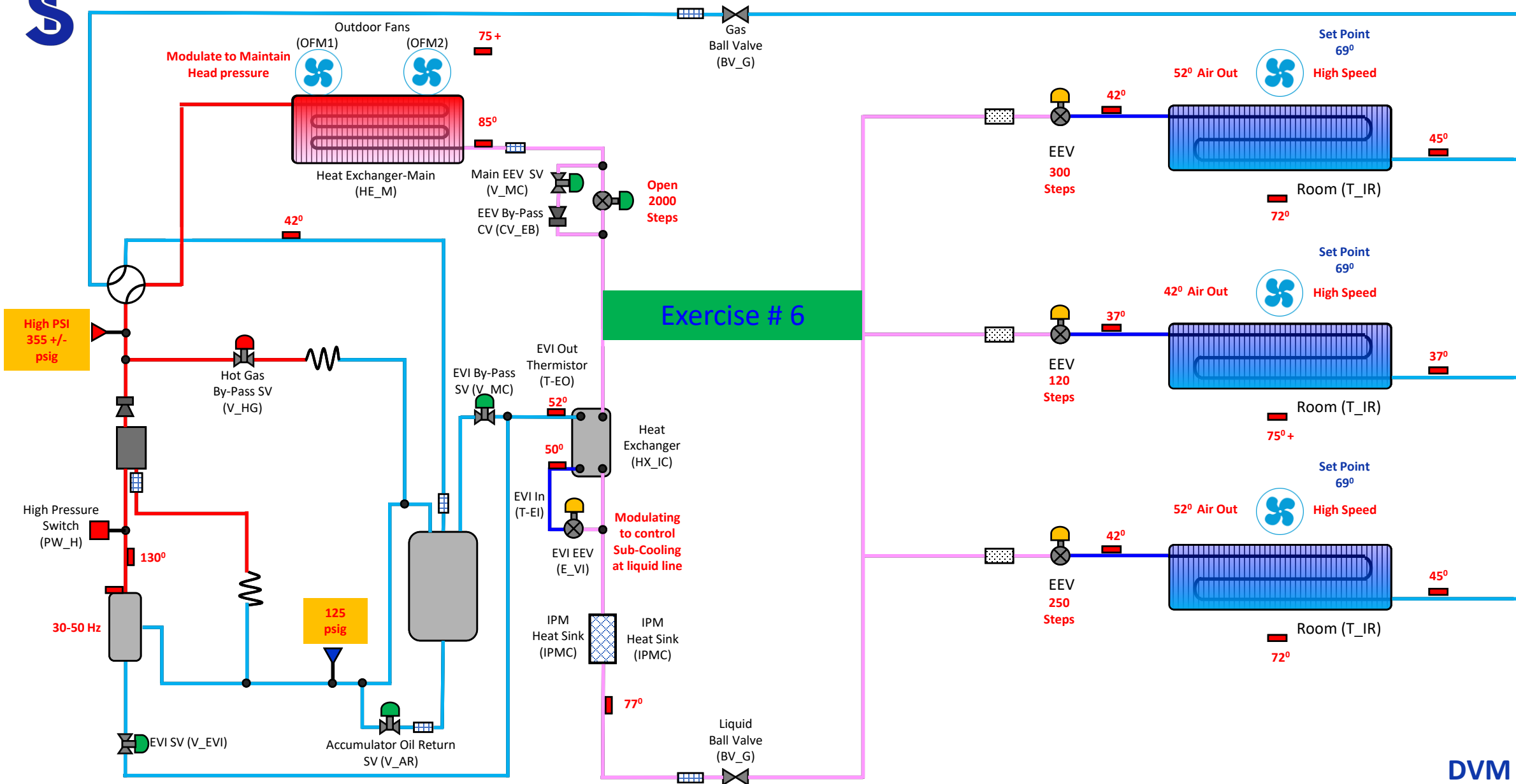
Exercise # 6

Operational Characteristics

- ☐ Inadequate cooling zone 2
- ☐ Cold air is coming from the unit but the space temperature will not drop

We can use this section to put in operational characteristics of the system and any other data pertinent to the fault, like the compressor is in protection mode or system is throwing a fault. Thoughts





DVM S HP

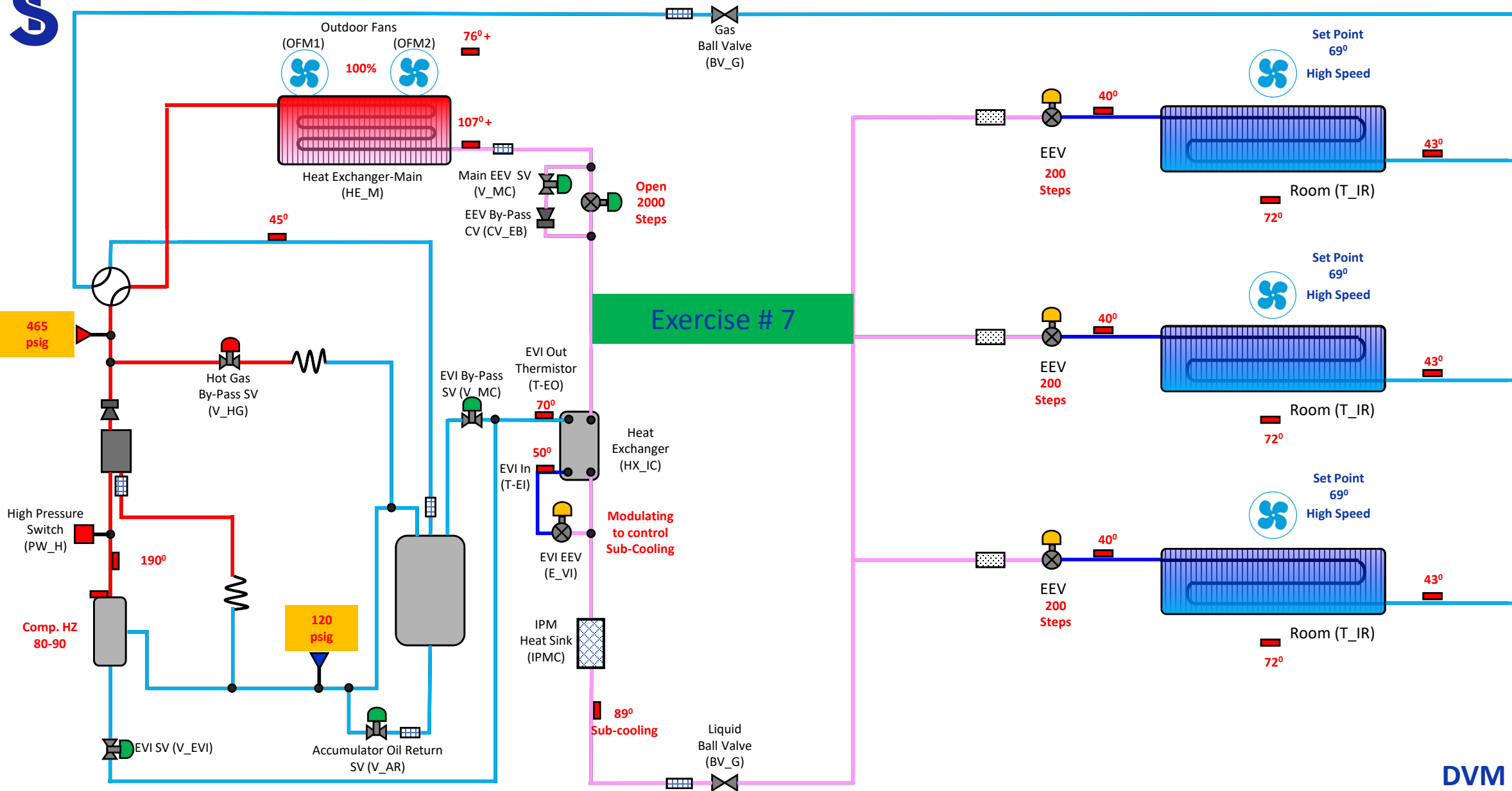
Exercise # 7

Operational Characteristics

- ☐ Inadequate cooling
- ☐ System Cools when the load is low but not when its high
- ☐ Compressor making noise

We can use this section to put in operational characteristics of the system and any other data pertinent to the fault, like the compressor is in protection mode or system is throwing a fault. Thoughts





DVM S HP

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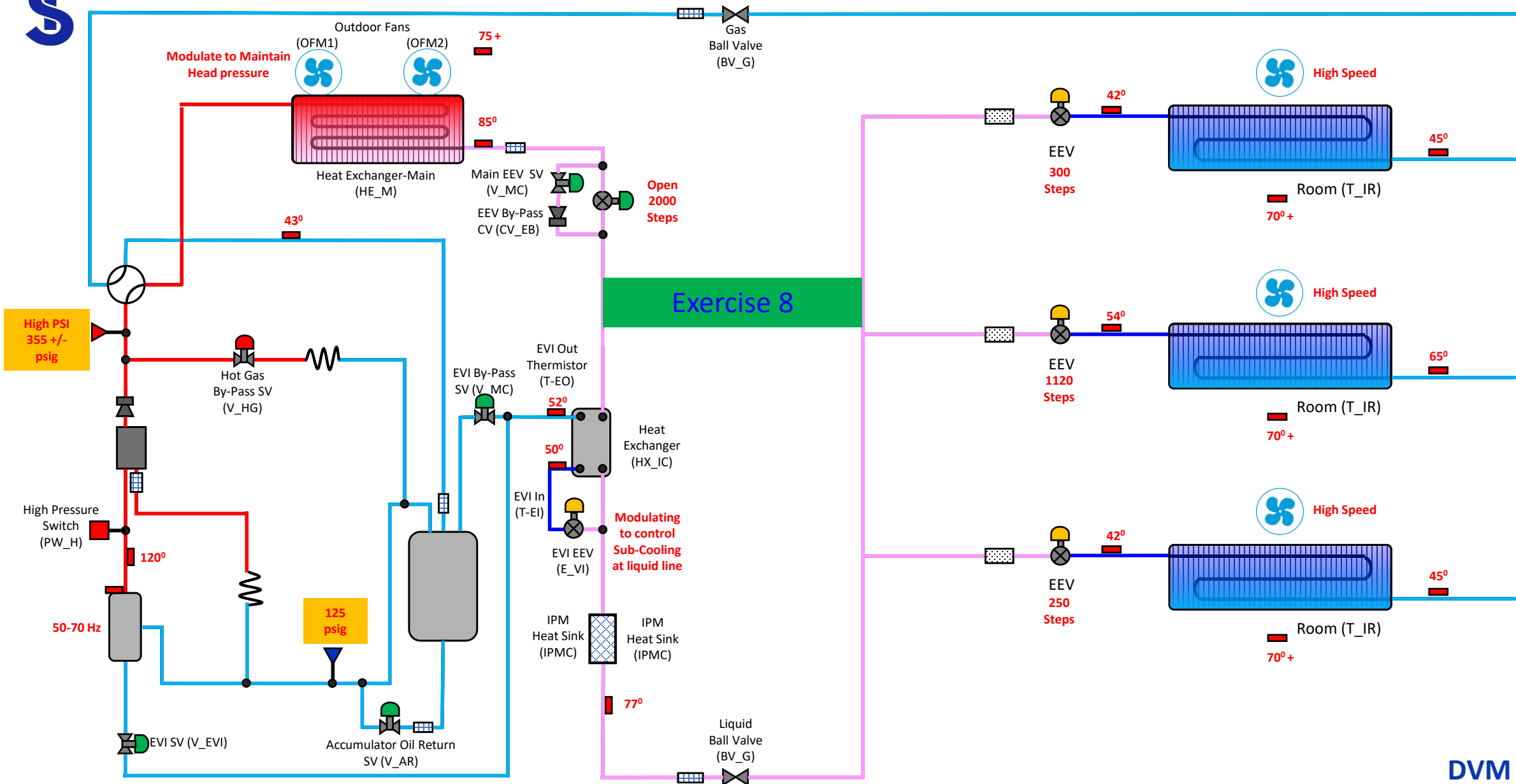
Exercise # 8

Operational Characteristics

- ☐ Zone 2 is not cooling
- ☐ Zones 1 & 3 have diminished cooling capacity

We can use this section to put in operational characteristics of the system and any other data pertinent to the fault, like the compressor is in protection mode or system is throwing a fault. Thoughts





DVM S HP

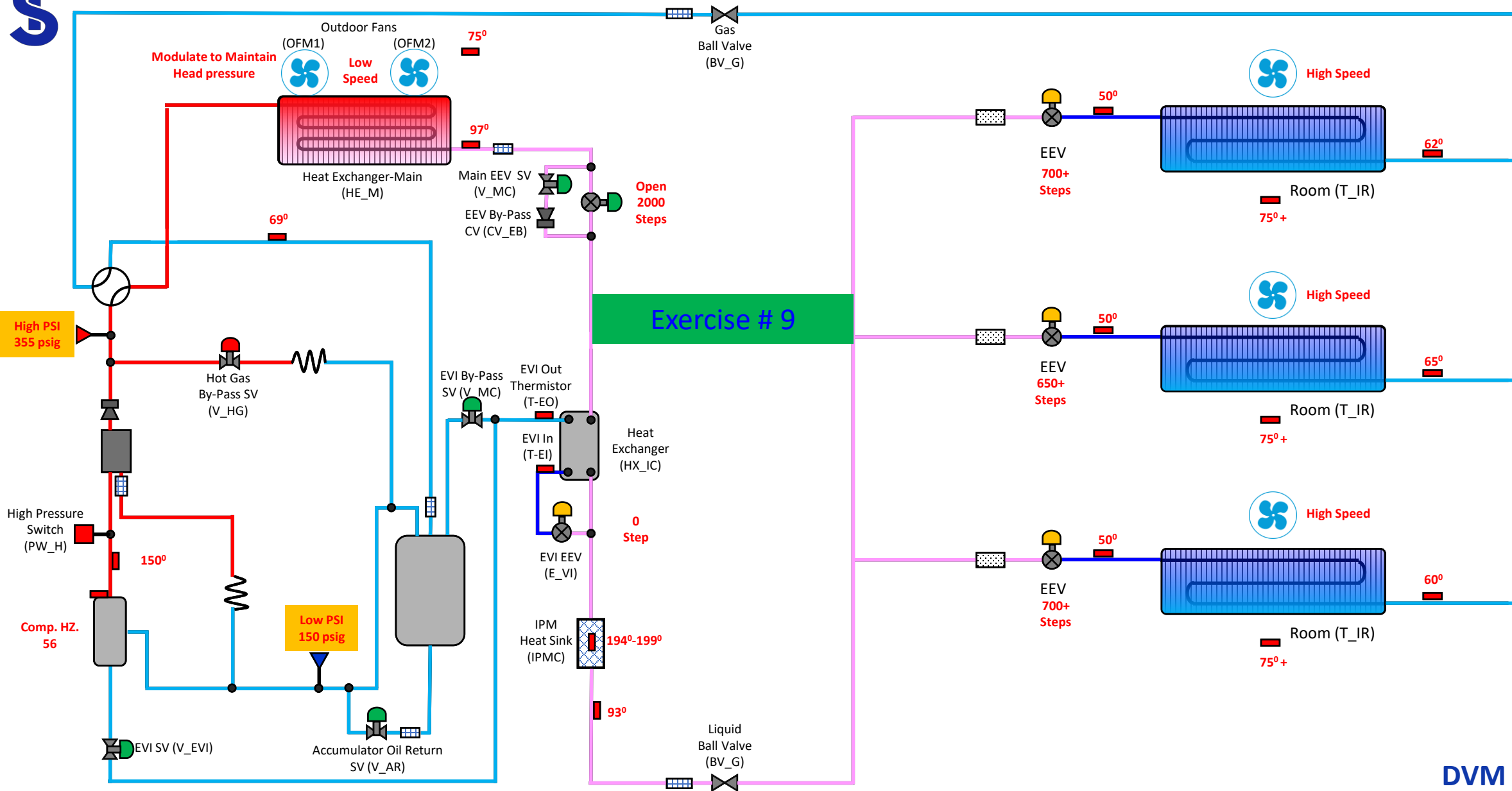
Exercise # 9

Operational Characteristics

- ☐ In adequate cooling
- ☐ Contractor had a bad compressor
- ☐ They replaced the compressor and IPM
- ☐ They verified the charge
- ☐ The compressor will not ramp up

We can use this section to put in operational characteristics of the system and any other data pertinent to the fault, like the compressor is in protection mode or system is throwing a fault. Thoughts





DVM S HP

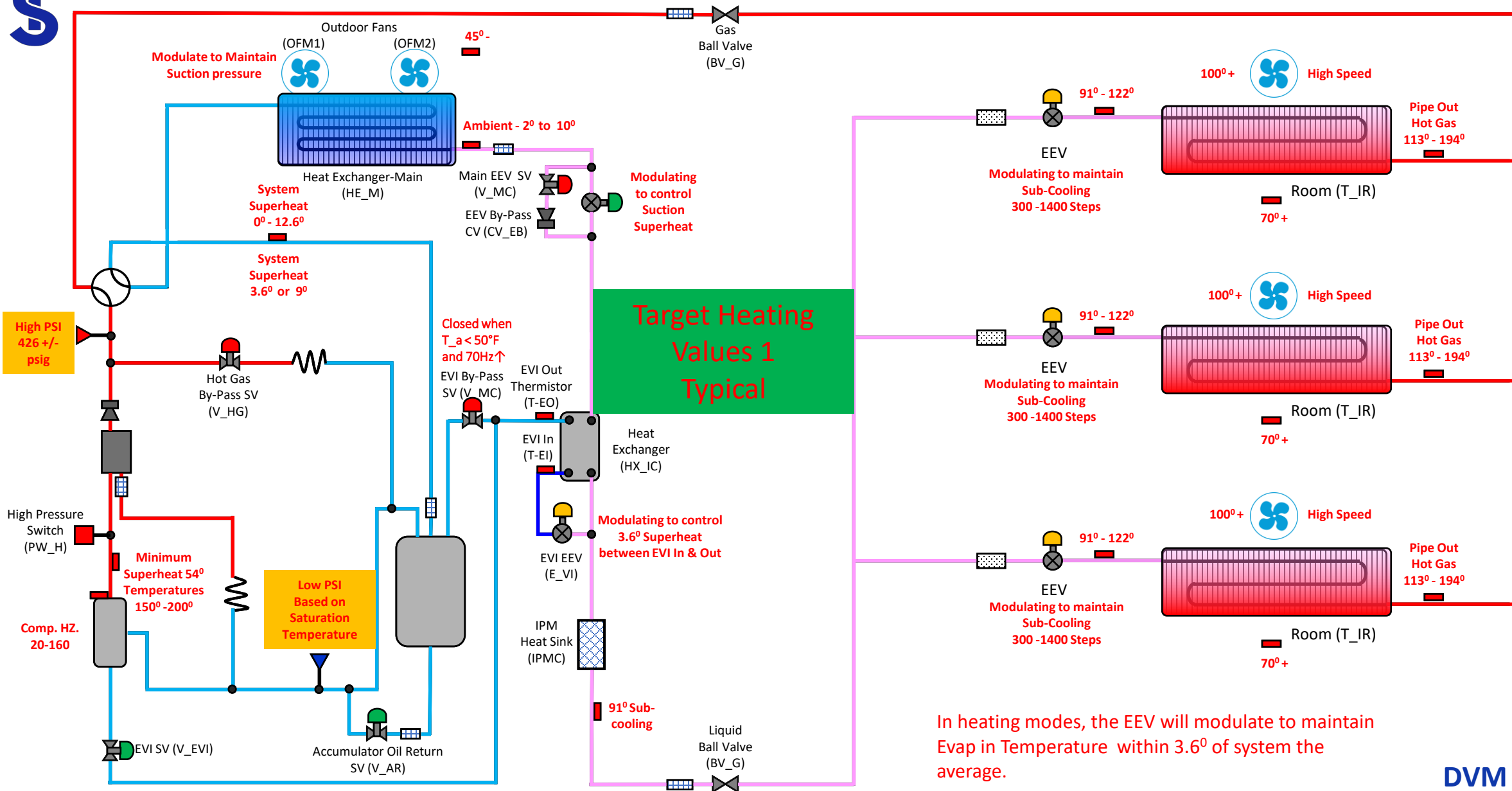
Heating Exercises



Check point of heat trial operation

※ The data is not absolute one, only can be used to check the abnormal status

Item	Normal	Check point
EVA out Temp.	113 ⁰ - 194 ⁰	▷ Normal condition of the 'Evap In' temperature - 23.4 ⁰ higher than room temp (14.4 ⁰ for duct model) - not less than 30.6 ⁰ from high pressure saturated temp (39.6 ⁰ for duct model) ▷ If all 'Evap In temp' is lower than normal(above) or the high pressure is lower than 355 psig, - Check whether the ambient temp is lower than 23 ⁰ / Load calculation for heating. - It can be caused by 'Refrigerant overcharged' or 'Wrong load design'. ▷ If only few indoor unit's Evap In temp. is lower than normal, check the pipe length from the first Y-joint to the indoor unit (limited 45m) also ensure that if there is any kind of flow resistance.
EVA in Temp.	91 ⁰ - 122 ⁰	
EEV Step of Indoor Unit	Initial 960 (RAC 1440) Higher floor ↓ Lower floor ↑ ODU closer ↓ ODU farther↑	
High Pressure psig	355 - 425 psig	▷ High pressure can be lower than normal in below case 1. Ambient temp 32 ↓ , 2. heating capacity shortage by wrong design, 3. Indoor temp 68 ↓ ▷ Low pressure will be a reference value because can be changed easily by ambient temperature
Low Pressure psig	71 – 106.6 psig	
Compressor Discharge Temp.	149 ⁰ - 221 ⁰	▷ If all indoor units are operating and both 1 and 2 is satisfied, it may be normal operation. 1. Highest DSH(discharge temp - high pressure saturated temp) ≥ 36 ⁰ 2. Max. discharge temp ≤ 221 ⁰ * Highest DSH<36: Ref. overcharged, EEV or liquid bypass valve or EVI EEV check * Max. dis temp>221: Ref. shortage, EEV malfunction, Liquid service valve close
Suction super heat	1.8 – 12.6	▷ Check EEV working if suction super heat is not satisfied. - If main eeV step is max. & super heat is high, then we can suspect refrigerant shortage
Main EEV step	300~1400	



In heating modes, the EEV will modulate to maintain Evap in Temperature within 3.6° of system the average.

DVM S HP

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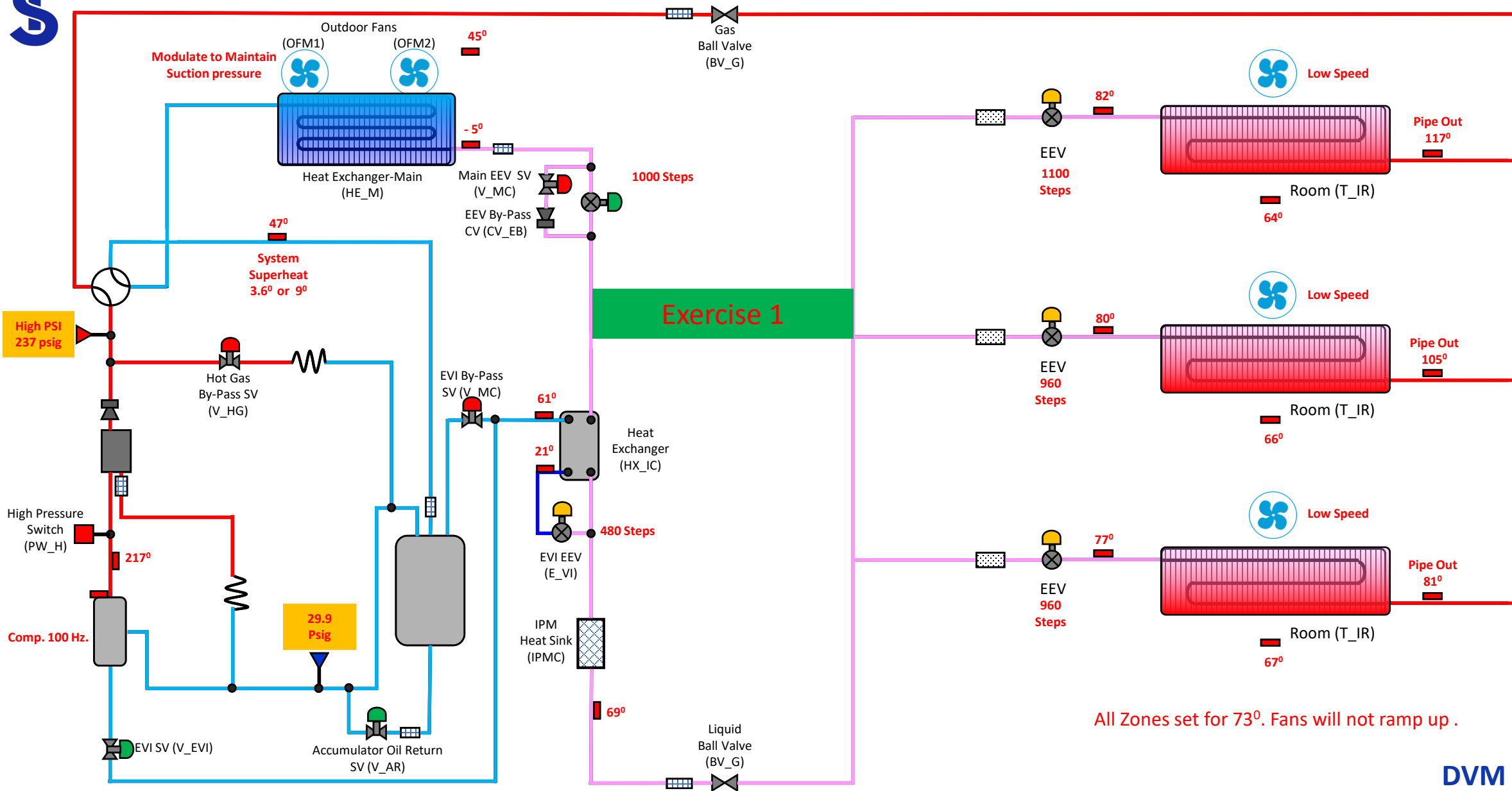
Exercise Heating #1

Operational Characteristics

- ❑ In adequate heating at full load
- ❑ Heating performance diminishes as outside air temperature drops.

We can use this section to put in operational characteristics of the system and any other data pertinent to the fault, like the compressor is in protection mode or system is throwing a fault. Thoughts





DVM S HP

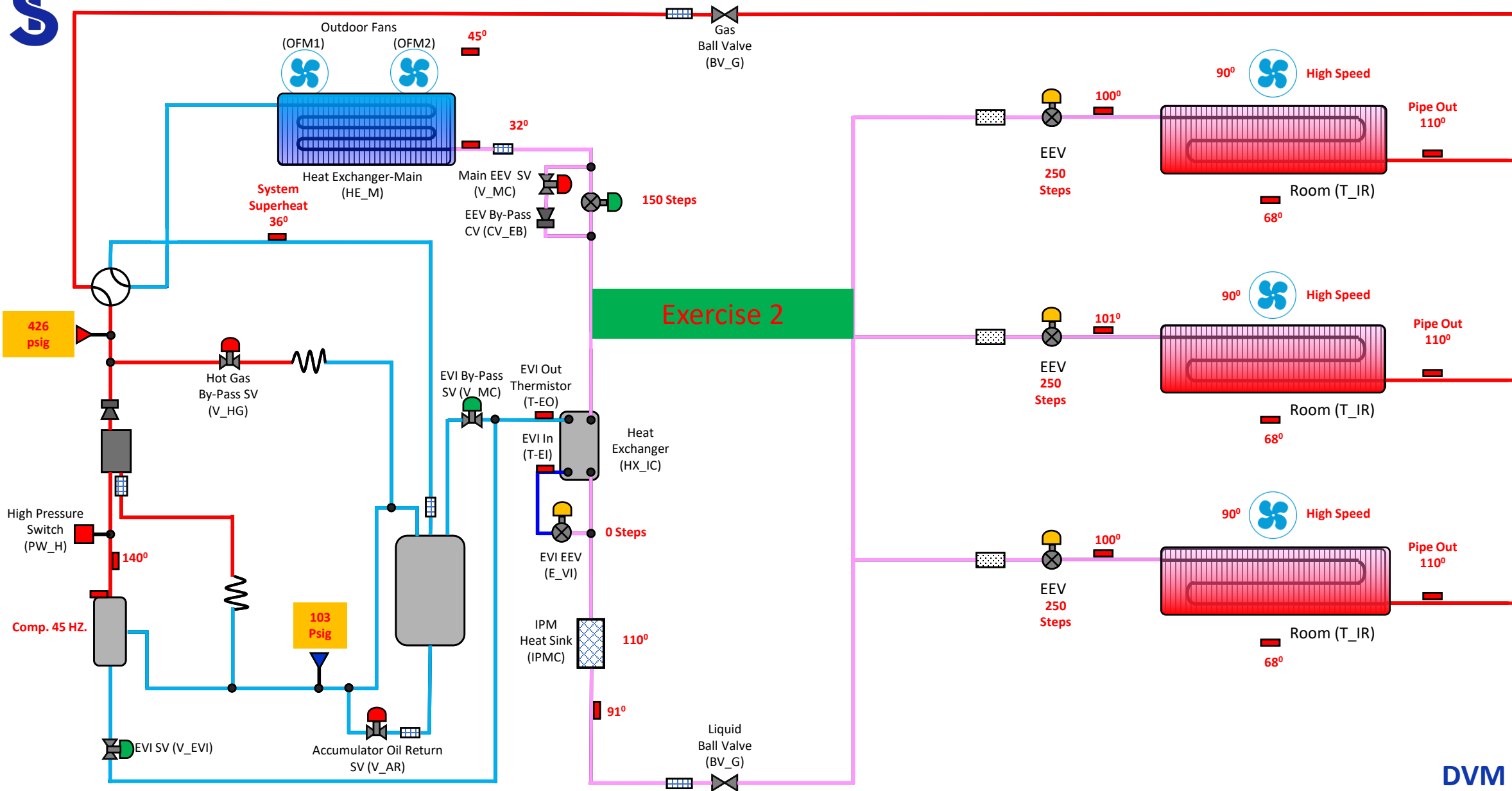
Exercise Heating #2

Operational Characteristics

- ☐ In adequate heating
- ☐ Compressor will not ramp up

We can use this section to put in operational characteristics of the system and any other data pertinent to the fault, like the compressor is in protection mode or system is throwing a fault. Thoughts





DVM S HP

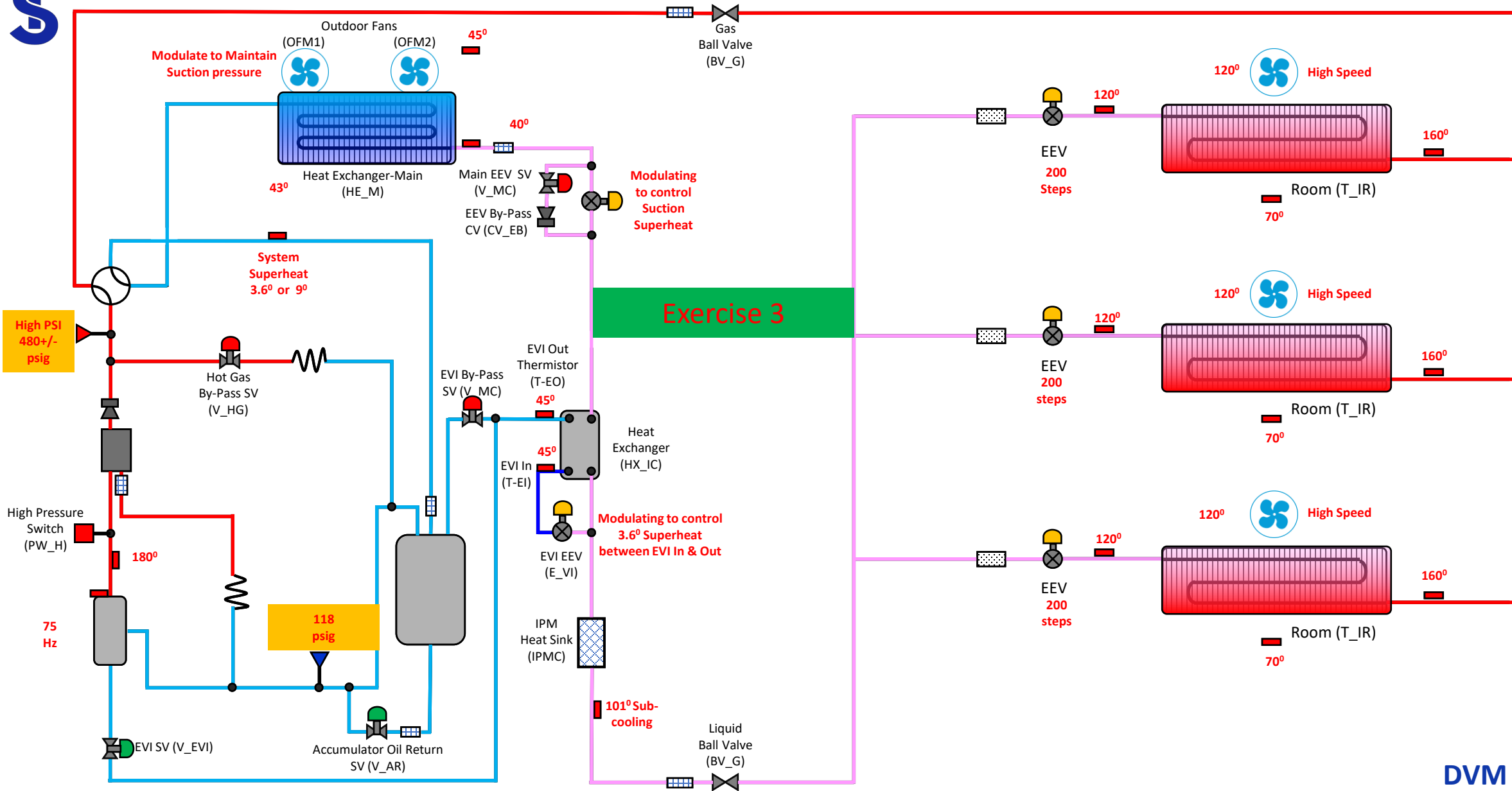
Exercise Heating #3

Operational Characteristics

- ☐ In adequate heating
- ☐ Compressor will not ramp up
- ☐ High TD's across IDU's
- ☐ Space temperature hot hitting setpoint

We can use this section to put in operational characteristics of the system and any other data pertinent to the fault, like the compressor is in protection mode or system is throwing a fault. Thoughts





DVM S HP

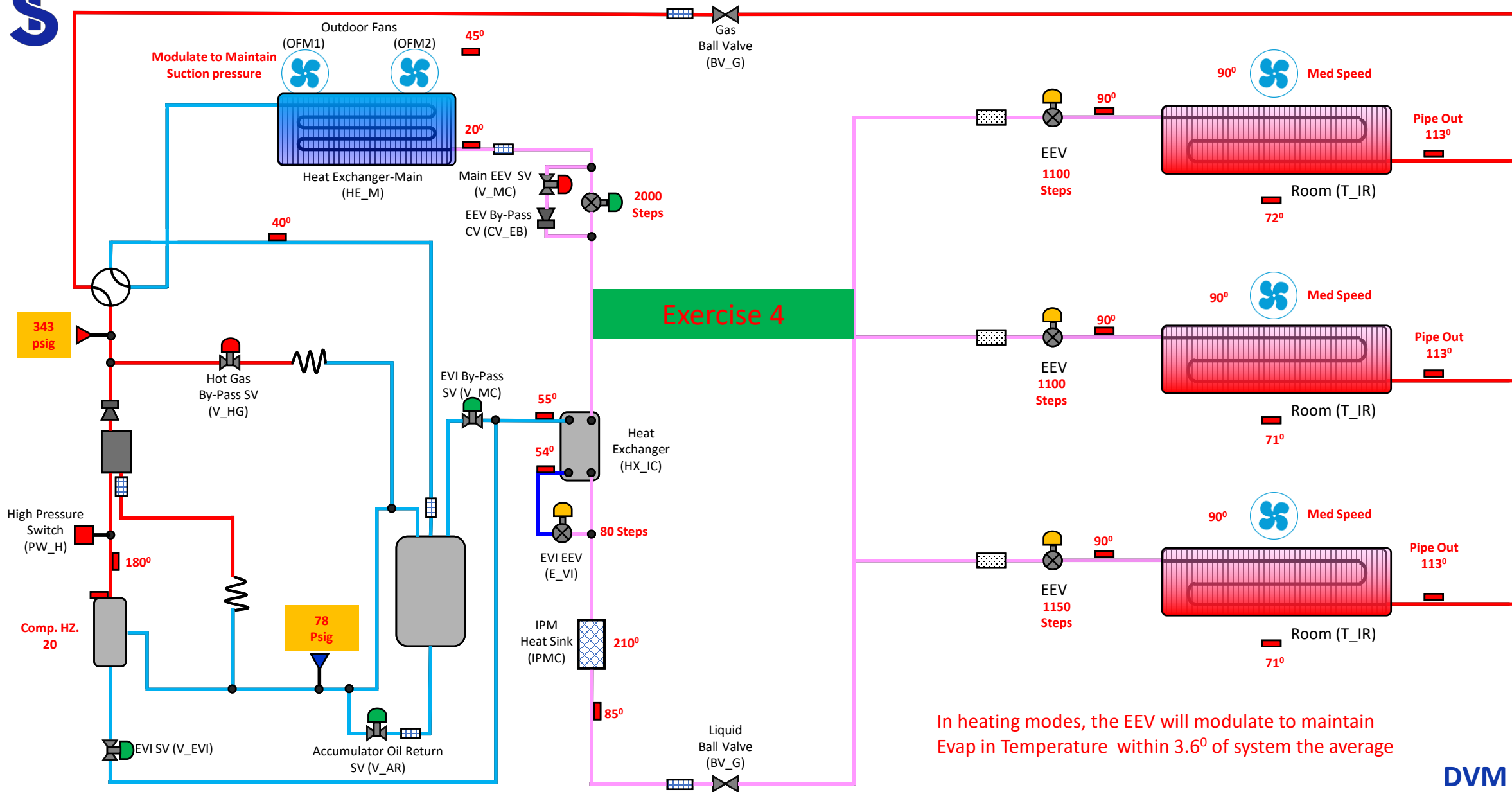
Exercise Heating #4

Operational Characteristics

- ☐ In adequate heating
- ☐ Compressor will not ramp up

We can use this section to put in operational characteristics of the system and any other data pertinent to the fault, like the compressor is in protection mode or system is throwing a fault. Thoughts





DVM S HP

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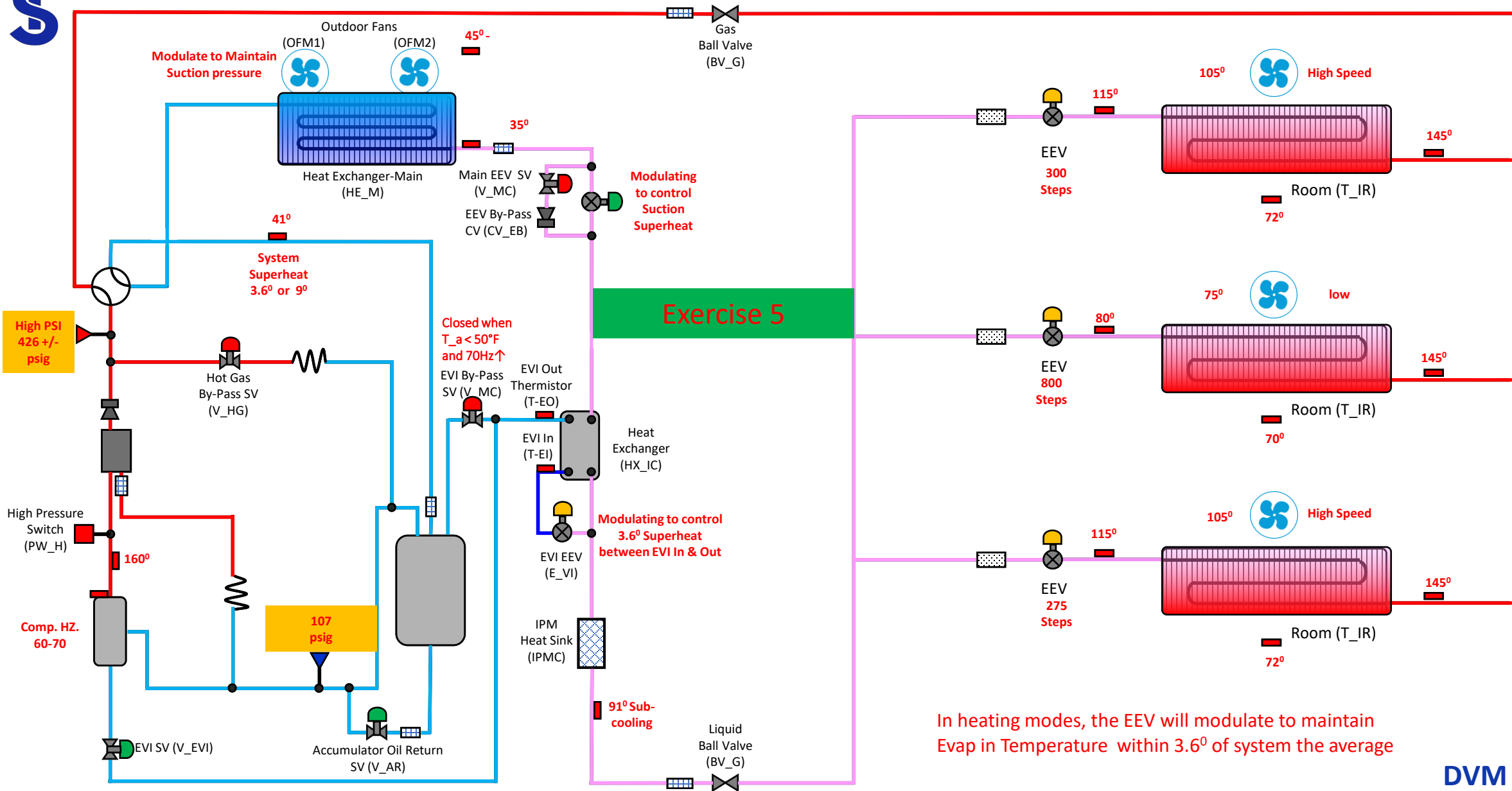
Exercise Heating #5

Operational Characteristics

- ☐ One zone not producing heat
- ☐ All other zones are working as required

We can use this section to put in operational characteristics of the system and any other data pertinent to the fault, like the compressor is in protection mode or system is throwing a fault. Thoughts





In heating modes, the EEV will modulate to maintain Evap in Temperature within 3.6° of system the average

DVM S HP

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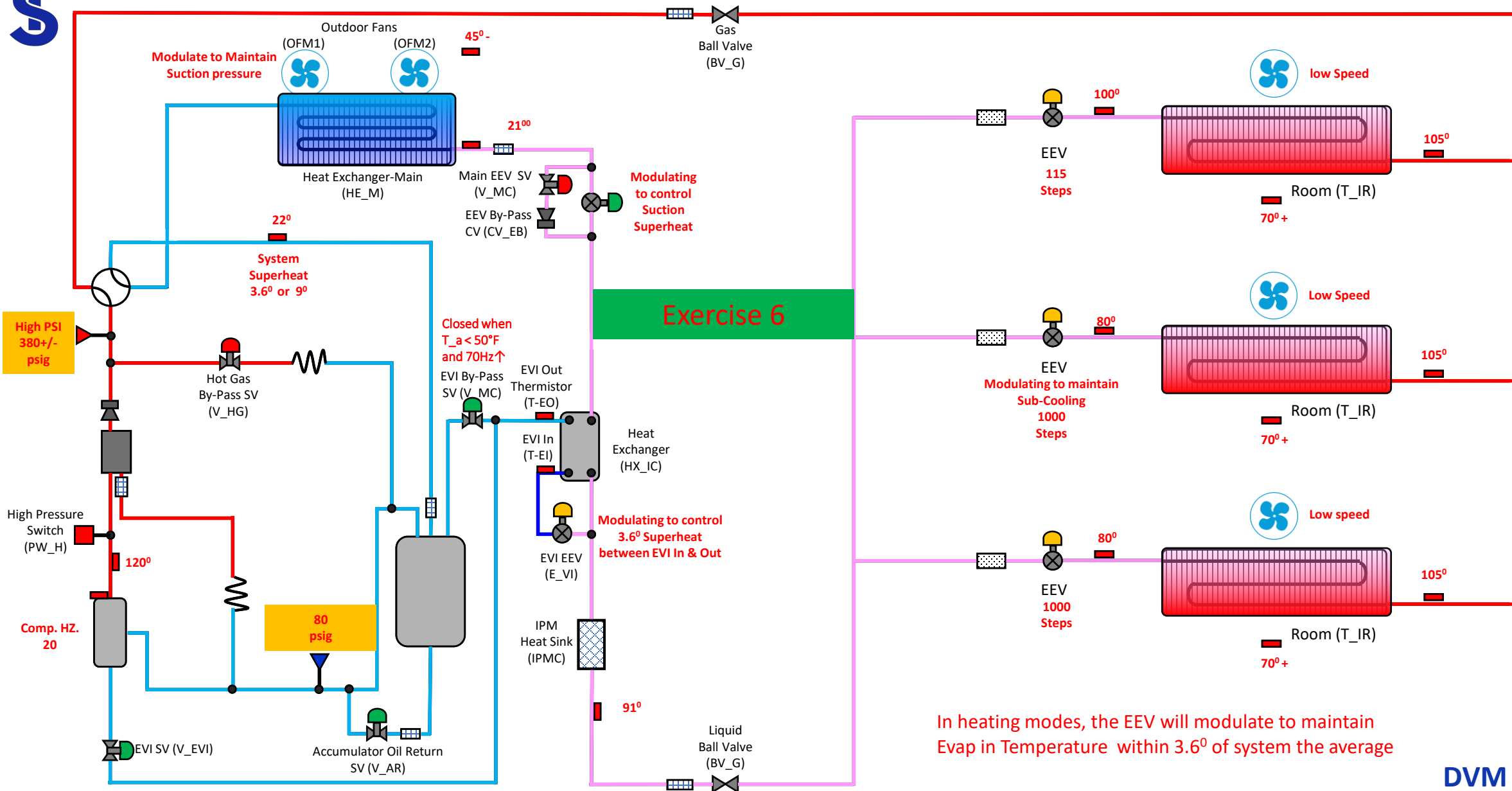
Exercise Heating #6

Operational Characteristics

- ☐ One zone not producing heat
- ☐ All other zones are working as required

We can use this section to put in operational characteristics of the system and any other data pertinent to the fault, like the compressor is in protection mode or system is throwing a fault. Thoughts





In heating modes, the EEV will modulate to maintain Evap in Temperature within 3.6° of system the average

DVM S HP

SAMSUNG

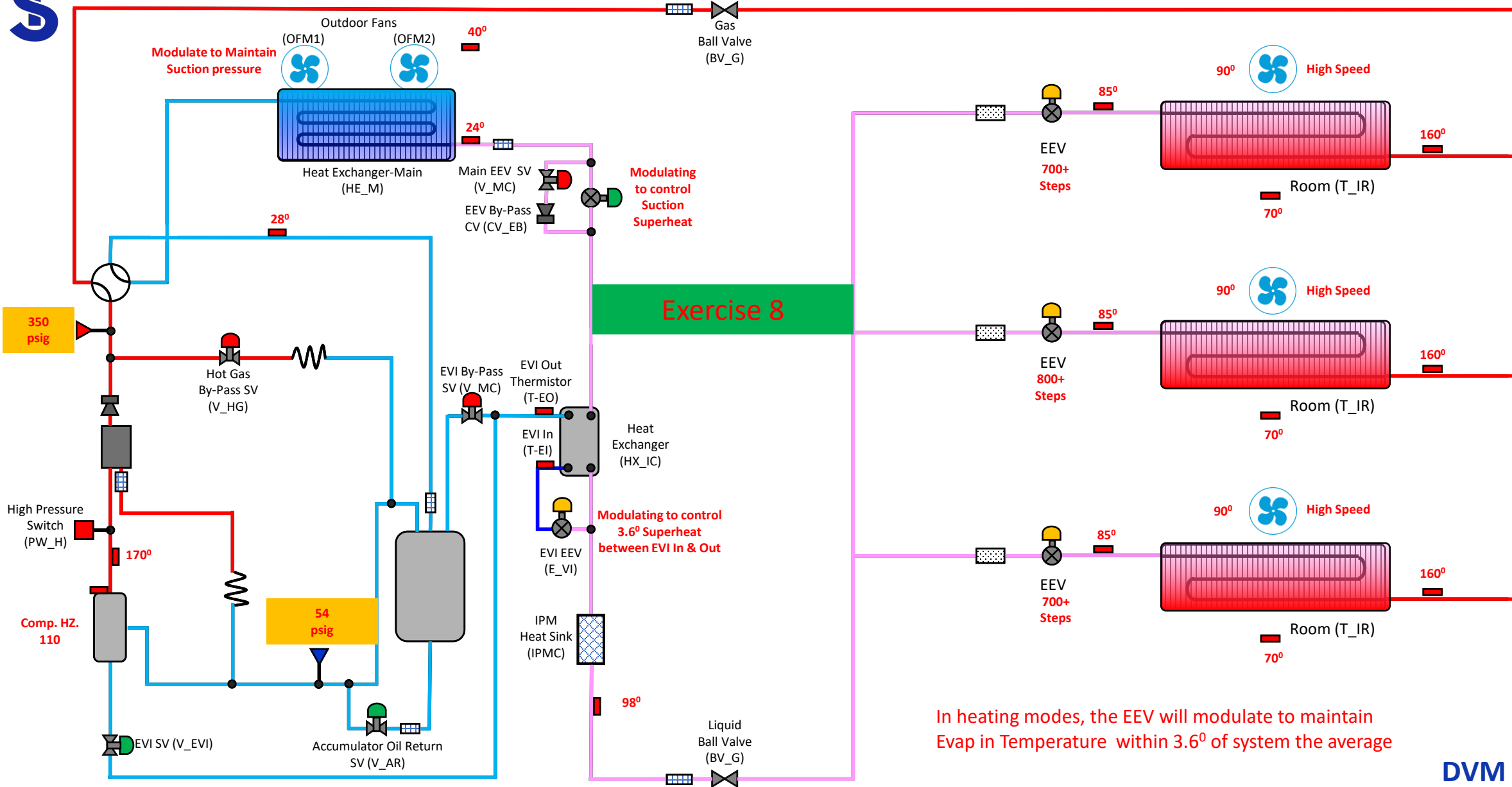
Exercise # 8

Operational Characteristics

- ❑ Erratic heating
- ❑ System heats under partial load and performance dies off as it get colder out and the interior load increases
- ❑ When all IDU's are calling with high fan speed the IDU Td's are small, and spaces will not hit setpoint

We can use this section to put in operational characteristics of the system and any other data pertinent to the fault, like the compressor is in protection mode or system is throwing a fault. Thoughts





DVM S HP